Chemical compass model of avian magnetoreception

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Supplementary Information

1. Supplementary Figures

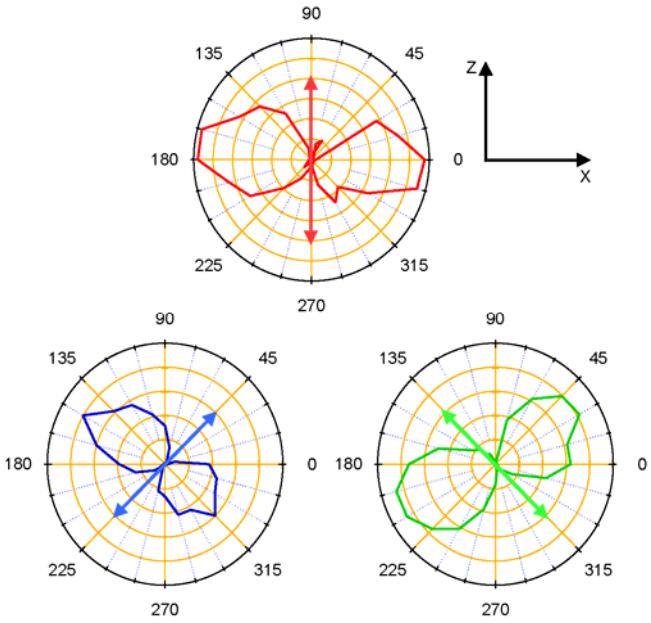


Fig. S1. Polar plots of the anisotropy of the magnetic field effect on the transient absorption of $[C^{\bullet+}PF^{\bullet-}]$ detected by photoselection (cf. Fig. 3). The experimental arrangement was as shown in Fig. 3a except that both the magnetic field and the polarization axis of the probe light (indicated here by the double headed arrows) lay in the *XZ* plane.

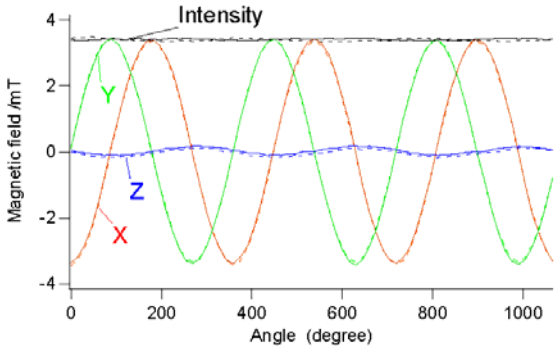


Fig. S2. Magnetic field calibration. See Methods for details.

2. Supplementary Calculations

Calculation of $C^{\bullet+}$ absorption bands

DFT calculations (Gaussian 03 [1]) of the optical transitions in $C^{\bullet+}$ (including both benzene rings and the amide bond between them) gave the following results (where *f* is the oscillator strength):

Method	Most prominent optical transitions
TD-DFT B3LYP 6-31G(d,p)	881 nm (*f* = 1.01) 821 nm (*f* = 1.62) 808 nm (*f* = 1.28)
TD-DFT B3LYP cc-pVTZ	876 nm (*f* = 1.23) 822 nm (*f* = 2.33) 739 nm (*f* = 0.40)
TD-DFT mPW1PW91 cc-pVTZ in PCM water	967 nm (*f* = 4.41)

The transition dipole moments for all the transitions listed here are very nearly aligned along the long axis of the carotenoid chain (Fig. S3):

Transition	Transition electric dipole moment vector (*X*, *Y*, *Z* / a.u.) (TD-DFT mPW1PW91 cc-pVTZ in PCM water)		
967 nm (*f* = 4.41)	–11.7856	–1.2100	0.2502

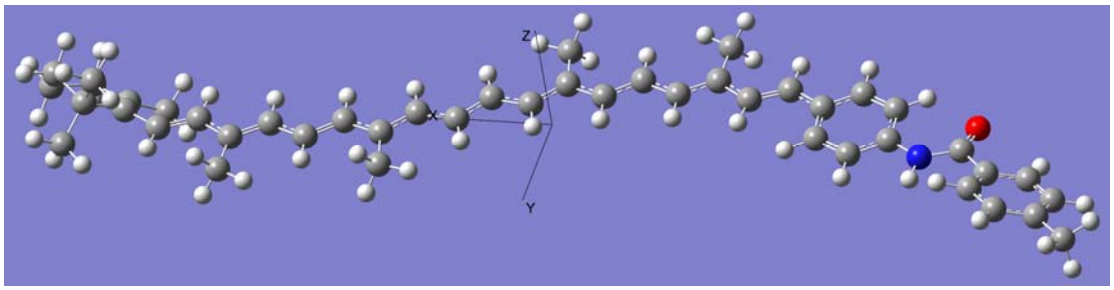


Fig. S3. The molecular coordinate system of the carotenoid moiety. The *XY* plane is the plane of the conjugated system.

[1] M. J. Frisch; G. W. Trucks; H. B. Schlegel; G. E. Scuseria; M. A. Robb; J. R. Cheeseman; J. A. Montgomery; Jr., T. V.; K. N. Kudin; J. C. Burant; J. M. Millam; S. S. Iyengar; J. Tomasi; V. Barone; B. Mennucci; M. Cossi; G. Scalmani; N. Rega; G. A. Petersson; H. Nakatsuji; M. Hada; M. Ehara; K. Toyota; R. Fukuda; J. Hasegawa; M. Ishida; T. Nakajima; Y. Honda; O. Kitao; H. Nakai; M. Klene; X. Li; J. E. Knox; H. P. Hratchian; J. B. Cross; C. Adamo; J. Jaramillo; R. Gomperts; R. E. Stratmann; O. Yazyev; A. J. Austin; R. Cammi; C. Pomelli; J. W. Ochterski; P. Y. Ayala; K. Morokuma; G. A. Voth; P. Salvador; J. J. Dannenberg; V. G. Zakrzewski; S. Dapprich; A. D. Daniels; M. C. Strain; O. Farkas; D. K. Malick; A. D. Rabuck; K. Raghavachari; J. B. Foresman; J. V. Ortiz; Q. Cui; A. G. Baboul; S. Clifford; J. Cioslowski; B. B. Stefanov; G. Liu; A. Liashenko; P. Piskorz; I. Komaromi; R. L. Martin; D. J. Fox; T. Keith; M. A. Al-Laham; C. Y. Peng; A. Nanayakkara; M. Challacombe; P. M. W. Gill; B. Johnson; W. Chen; M. W. Wong; C. Gonzalez; Pople, J. A. *Gaussian 03, Revision D.02*, 2004.

3. Supplementary theory

Orientational averaging of the magnetic field effect on the CPF triad In the following, we relate the anisotropic signals measured in the liquid crystal and the photoselection experiments to the underlying radical pair magnetic field effect by considering the orientational averaging inherent in the two approaches.

We define the molecular frame as the coordinate system in which the transition dipole moment of $C^{\bullet+}$ in $[C^{\bullet+}P\,F^{\bullet-}]$ lies along the *z*-axis. Note that this axis system, chosen here for convenience, differs from that used in calculating the direction of the transition dipole moment (Section S1.3). The laboratory frame is defined such that the applied magnetic field $\tilde{\mathbf{B}}$, where the tilde denotes a vector in the laboratory frame, is directed along the *z*-axis and the polarization of the probe beam $\tilde{\mathbf{p}}$ is in the *yz*-plane

$$\tilde{\mathbf{B}} = \begin{pmatrix} 0 \\ 0 \\ B \end{pmatrix}; \qquad \tilde{\mathbf{p}} = \begin{pmatrix} 0 \\ \sin\varepsilon \\ \cos\varepsilon \end{pmatrix} \tag{1}$$

where ε is the angle between $\tilde{\mathbf{p}}$ and $\tilde{\mathbf{B}}$. Note that, although geometrically equivalent, the (*x*, *y*, *z*) laboratory axis system used here differs from that used to describe the actual experiment (Fig. 3a).

Photoselection experiment The molecular and laboratory frames are related by a rotation, which may be parameterised by the Euler angles (α, β, γ). We deal first with isotropic samples, where photoselection is used to measure the anisotropic response to the applied magnetic field, so that all rotations are equally probable. Rotation by the Euler angles may be written in matrix form:

$$\begin{aligned} \mathsf{R} &= \mathsf{R}_z(\gamma)\mathsf{R}_{-y}(\beta)\mathsf{R}_z(\alpha) \\ &= \begin{pmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{pmatrix} \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned} \tag{2}$$

In the molecular frame, the magnetic field is described by the vector

$$\mathbf{B} = \mathsf{R}\tilde{\mathbf{B}} = \begin{pmatrix} B\sin\beta\cos\gamma \\ B\sin\beta\sin\gamma \\ B\cos\beta \end{pmatrix} \tag{3}$$

which may be written in spherical polar coordinates as (B, β, γ). In the molecular frame, the probe polarization vector **p** is given by

$$\mathbf{p} = \mathsf{R}\tilde{\mathbf{p}} = \begin{pmatrix} \cos\varepsilon\cos\gamma\sin\beta - \cos\beta\cos\gamma\sin\alpha\sin\varepsilon - \cos\alpha\sin\varepsilon\sin\gamma \\ \cos\alpha\cos\gamma\sin\varepsilon - \cos\beta\sin\alpha\sin\gamma\sin\varepsilon + \cos\varepsilon\sin\beta\sin\gamma \\ \cos\beta\cos\varepsilon + \sin\alpha\sin\beta\sin\varepsilon \end{pmatrix} \tag{4}$$

The probability that a molecule with orientation (α, β, γ) will be detected is

$$P_{\mathrm{ex}}(\alpha, \beta, \gamma, \varepsilon) \propto \left|\mathbf{k}\cdot\mathbf{p}\right|^2 \tag{5}$$

where **k** is the unit vector along the molecular frame z-axis (i.e. the direction of the transition dipole moment) and the constant of proportionality is chosen such that

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{2\pi} P_{\mathrm{ex}}(\alpha, \beta, \gamma, \varepsilon)\sin\beta \,\mathrm{d}\alpha \,\mathrm{d}\beta \,\mathrm{d}\gamma = 1 \tag{6}$$

The orientationally averaged singlet yield (i.e. the relative yield of the product formed by spin-selective recombination of the singlet state of the radical pair) is

$$\left\langle \Phi_{\mathrm{S}} \right\rangle (B, \varepsilon) = \int_0^{2\pi}\int_0^{\pi}\int_0^{2\pi} P_{\mathrm{ex}}(\alpha, \beta, \gamma, \varepsilon)\, \Phi_{\mathrm{S}}(B, \beta, \gamma)\sin\beta \,\mathrm{d}\alpha \,\mathrm{d}\beta \,\mathrm{d}\gamma \tag{7}$$

where $\Phi_{\mathrm{S}}(B, \beta, \gamma)$ is the singlet yield in a magnetic field **B** = (B, β, γ). Although we discuss the product yield here, we expect essentially identical results for the radical pair lifetime. In general, $\Phi_{\mathrm{S}}(B, \beta, \gamma)$ may be written as a sum of spherical harmonics with field-dependent coefficients, $a_{lm}(B)$:

$$\Phi_S(B,\beta,\gamma) = \sum_{l=0,2,4,\ldots} \sum_{m=-l}^{l} a_{lm}(B) Y_l^m(\beta,\gamma) \tag{8}$$

Note that the sum runs over even values of l because the singlet yield is invariant to inversion of the magnetic field ($\Phi_S(B,\beta+\pi,\gamma) = \Phi_S(B,\beta,\gamma)$).

Combining Eqns (4), (5) and (7) and integrating with respect to α we obtain

$$\begin{aligned} &\langle \Phi_S \rangle (B,\varepsilon) \\ &\quad = \frac{3}{8\pi} \int_0^{2\pi} \int_0^{\pi} \left[\sin^2\varepsilon + \cos^2\beta (3\cos^2\varepsilon - 1) \right] \Phi_S(B,\beta,\gamma) \sin\beta \, \mathrm{d}\beta \, \mathrm{d}\gamma \end{aligned} \tag{9}$$

Noting that

$$\frac{3}{8\pi}\left[\sin^2\varepsilon + \cos^2\beta (3\cos^2\varepsilon - 1) \right] = \frac{1}{2\sqrt{\pi}} Y_0^0(\beta,\gamma) + \frac{1}{2\sqrt{5\pi}} \left[3\cos^2\varepsilon - 1 \right] Y_2^0(\beta,\gamma) \tag{10}$$

and using

$$\int_0^{2\pi} \int_0^{\pi} \left[Y_l^m(\beta,\gamma) \right]^* Y_{l'}^{m'}(\beta,\gamma) \sin\beta \, \mathrm{d}\beta \, \mathrm{d}\gamma = \delta_{ll'} \delta_{mm'} \tag{11}$$

$$\left[Y_l^m(\beta,\gamma) \right]^* = (-1)^m Y_l^{-m}(\beta,\gamma) \tag{12}$$

we find:

$$\langle \Phi_S \rangle (B,\varepsilon) = \frac{1}{2\sqrt{\pi}} a_{00}(B) + \frac{1}{2\sqrt{5\pi}} \left[3\cos^2\varepsilon - 1 \right] a_{20}(B) \tag{13}$$

Thus, the orientational averaging inherent in the photoselection experiment leads to a dependence of the product yield on the direction of the probe polarization axis, ε, of the form

$$\langle \Phi_S \rangle(B,\varepsilon) = b_0 + b_2 \sin^2 \varepsilon \tag{14}$$

Liquid crystal experiment The orientational averaging for the triad molecule oriented in a liquid crystal is more straightforward. We assume that the molecular *z*-axis (the long axis of the molecule) is perfectly aligned with liquid crystal director and that the molecular *x*- and *y*-axes are randomly distributed. β is the angle between the molecular *z*-axis and the magnetic field direction, and γ specifies the rotation of the molecule around the *z*-axis. Averaging over γ gives

$$\langle \Phi_S \rangle(B,\beta) = \frac{1}{2\pi}\int_0^{2\pi} \Phi_S(B,\beta,\gamma)\,\mathrm{d}\gamma = \frac{1}{2\pi}\sum_{l=0,2,4,\ldots}\sum_{m=-l}^{l}\int_0^{2\pi} a_{lm}(B) Y_l^m(\beta,\gamma)\,\mathrm{d}\gamma \tag{15}$$

and hence

$$\langle \Phi_S \rangle(B,\beta) = \sum_{l=0,2,4,\ldots} a_{l0}(B)\sqrt{\frac{2l+1}{4\pi}} P_l^0(\cos\beta) \tag{16}$$

in which $P_l^0(x)$ is the associated Legendre polynomial. Since $P_l^0(x)$ is a polynomial in *x* with powers up to and including x^l, $\langle \Phi_S \rangle(B,\beta)$ has the form

$$\langle \Phi_S \rangle(B,\beta) = \sum_{l=0,2,4,\ldots} c_l \sin^l \beta \tag{17}$$

By comparison with Eqn (14), it is clear that the orientational averaging of $\langle \Phi_S \rangle(B,\beta)$ here is less extensive than it is for $\langle \Phi_S \rangle(B,\varepsilon)$ in the photoselection measurement. However, if the anisotropy of the magnetic field effect is well described by second-order spherical harmonics, i.e. if $a_{l0}(B) = 0$ for $l > 2$, then

$$\langle \Phi_S \rangle(B,\beta) = c_0 + c_2 \sin^2 \beta \tag{18}$$

and $\langle \Phi_S \rangle (B, \beta)$ shows the same dependence on β as does $\langle \Phi_S \rangle (B, \varepsilon)$ on ε, as found experimentally (Fig. 3).